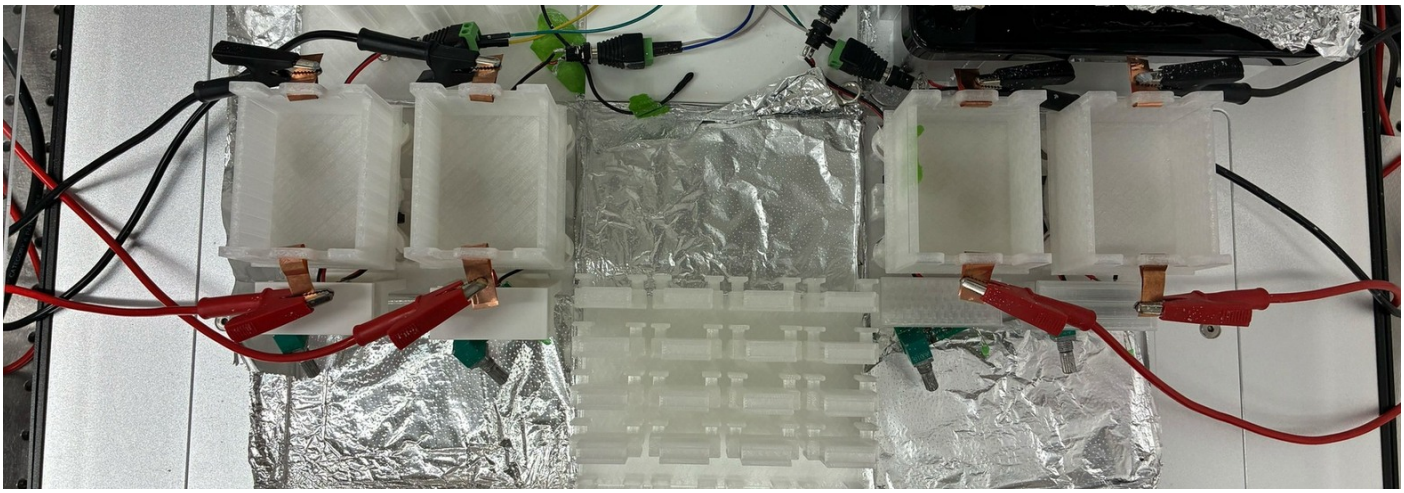
Guohan (Harry) Zhang

Engineering Portfolio

Johns Hopkins University | Electrical Engineering
Applied Mathematics minor | May 2028 | GPA 3.8/4.0

Selected work for Summer 2027 internships in manufacturing, robotics, semiconductor equipment, electrical engineering and applied AI.



Detail of the Luo Lab electrodeposition setup. Laboratory photograph.

Projects

Luo Lab 02–03
Robot integration, fixture design, XRD and DFT

CMOS circuits 04
Digital hierarchy, current-mode banks and circuit simulation

Madrid / ICAI 05
Neural-network boundary sampling and topology analysis

Scrawl AI 06–07
Personal handwriting extraction, rendering engine and web/iOS apps

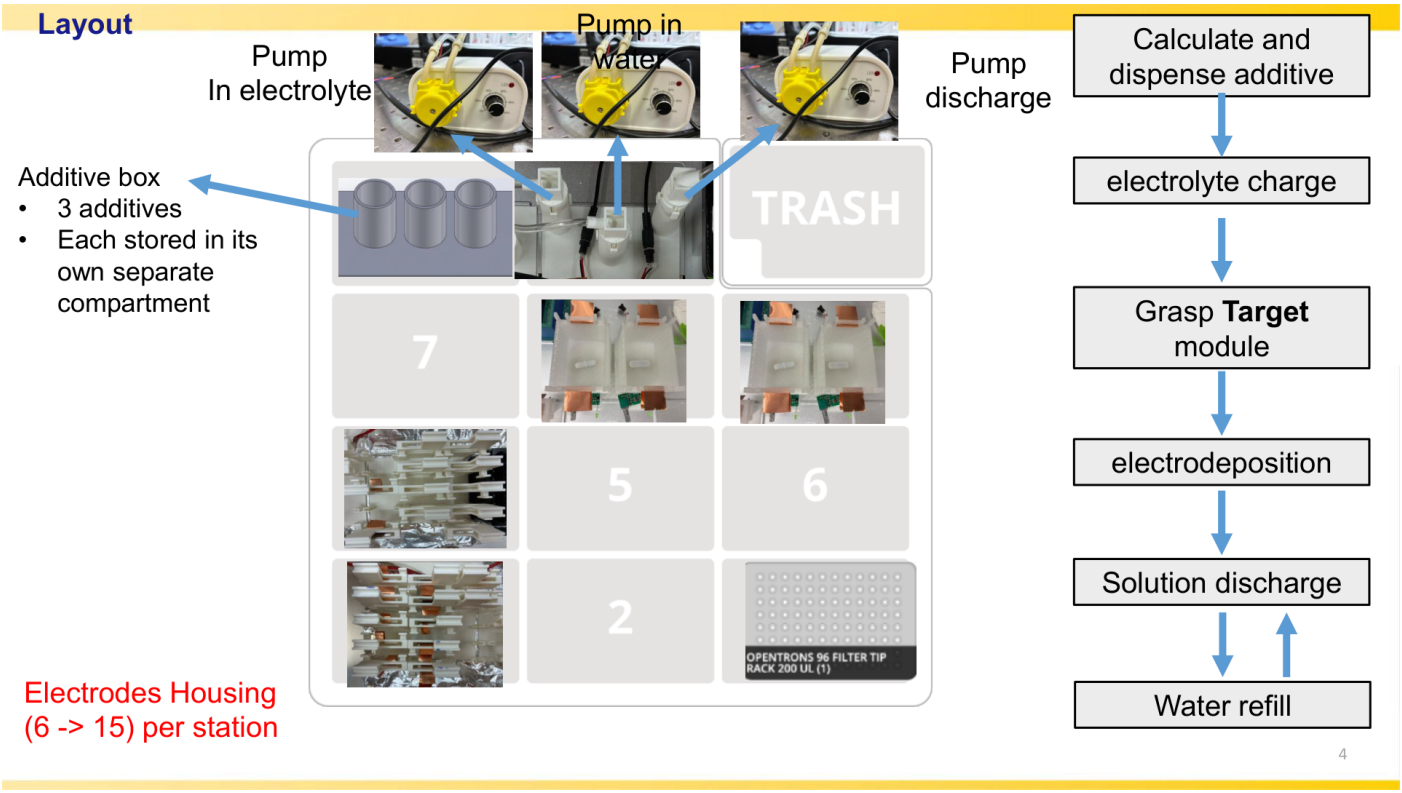
Costream 08
Python backend, data pipelines and enterprise onboarding

gzhang45@jh.edu linkedin.com/in/harrygzhang

Zinc electrodeposition automation

Undergraduate Researcher | Johns Hopkins University | Spring 2025–Present

Automated zinc-electrodeposition experiments using an OT-2 robot, NI DAQ hardware, ultrasonic actuation and custom fixtures.



Team setup and experiment sequence from Luo Lab slides credited to Peilin Li. The figure shows additive handling, pumps, electrode storage and the deposition cycle. My contributions are listed below.

My work

Robot integration

Integrated OT-2 robotics with ultrasonic actuation.

Electrical control

Developed Python/NI DAQ routines for waveform generation and pressure-valve sequencing.

Fixture design

Designed and 3D-printed experimental fixtures and housings in Autodesk Fusion.

Results

300% higher synthesis throughput.

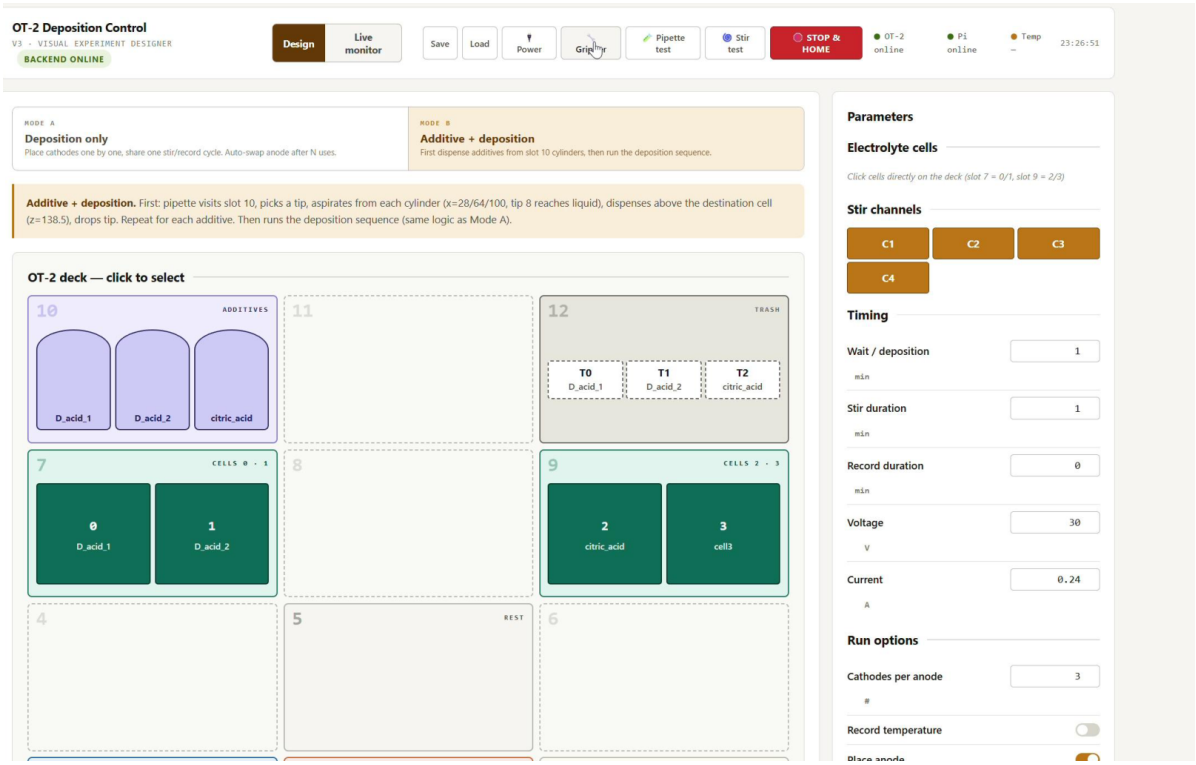
50+ experimental trials.

80% shorter hardware iteration lead time.

Lab controls, XRD and DFT

Luo Lab | Ralph O'Connor Sustainable Energy Institute (ROSEI)

Integrated laboratory controls, automated X-ray diffraction workflows and ran density functional theory calculations for zinc-ion battery research.



Team OT-2 control interface from Luo Lab slides credited to Peilin Li. It shows coordinated experiment settings and controls; my contributions are described below.

Automation and characterization

- Fully automated XRD characterization workflows.
- Integrated robot operation, instrument controls and experimental fixtures.

DFT calculations

- Investigated surface energies and additive adsorption across three zinc facets in VASP.
- Scheduled calculations on Rockfish HPC using Slurm.

Fixture design

Designed a driver-board enclosure and ultrasonic-transducer holder in Autodesk Fusion; planned OT-2 deck layouts.

Lab

Professor Yuting Luo's group studies zinc-ion batteries. Her 2026 NSF CAREER Award supports the research.

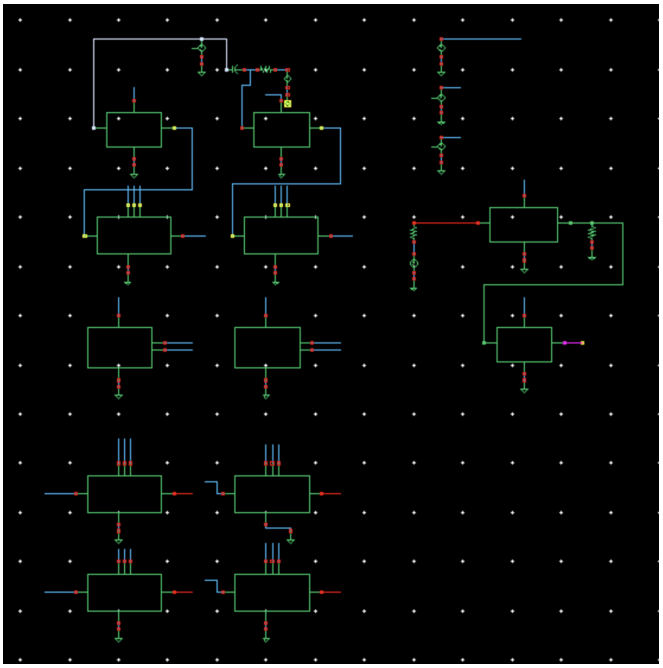
Related skills: Autodesk Fusion (CAD), 3D printing, Avizo, Arduino, NI DAQ, Python.

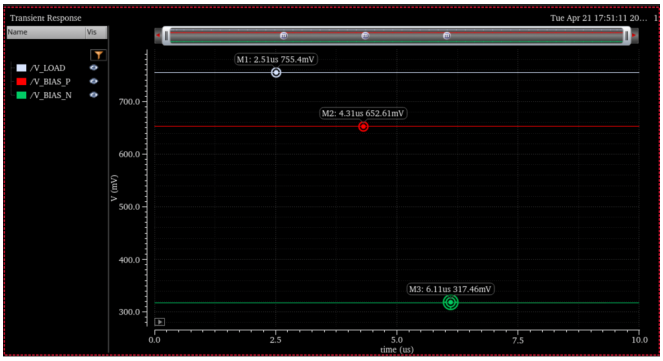
Luo Lab team

CMOS keyword-spotting circuits

Team course project | Introduction to VLSI | Spring 2026
Johns Hopkins University | With Andrew Chae

Designed and simulated mixed-signal circuit blocks for an audio keyword detector using Cadence Virtuoso, Spectre and the gpdk045 process kit.





Matched-pattern simulation of the 16-tap demonstration bank. The output settled near 755 mV.

Matched input	**755 mV**
Anti-matched input	**272 mV**
All taps active	**526 mV**

Team schematic from the final report: two shift-register paths feed four current-mode banks and threshold/detection stages. The full transient run was not completed.

My work

- Implemented transmission gates, inverters and flip-flops, then built two 1,024-stage shift registers.
- Built current-mode bank subcells, bias circuits, the threshold block and Week 3 integration testbench.
- Helped map MATLAB-derived weights to bias domains and debug integration.

Simulation and scope

- Ran matched, anti-matched and cancellation tests on the demonstration bank.
- The four-bank extension passed schematic and DC checks. Disk limits prevented a complete transient run.
- Schematic-level course work; no completed physical layout or fabricated chip.

Tools Cadence Virtuoso, Spectre, CMOS circuit design, hierarchical schematics, DC and transient simulation.

Neural-network topology

Research Student | Comillas Pontifical University (ICAI)
Madrid, Spain | May–Jul 2026

In a team study of neural-network topology, I developed an adaptive boundary search using Sobol sampling, Gaussian-process uncertainty and bisection.

Known geometries + synthetic data → **PyTorch training + checkpoints** → **Boundary extraction** → **Topology + diagnostics**

Connected components in stable boundary results

n = 892 stable, detected-boundary results



Stored team experiment: 1,000 random 2–8–1 ReLU networks. Of these, 892 had stable detected-boundary results, 100 had no detected boundary and 8 were resolution-sensitive.

This chart uses 2D grid-contour extraction on a 901 × 901 grid, checked against a 451 × 451 subsample, within $[-9, 9]^2$. It does not use the Gaussian-process sampler. Counts are from the stored experiment, not a new benchmark.

My work

- Developed an adaptive boundary search using Sobol sampling, Gaussian-process uncertainty and bisection.
- Contributed to PyTorch training and GUDHI persistent-homology analysis.

Limitations

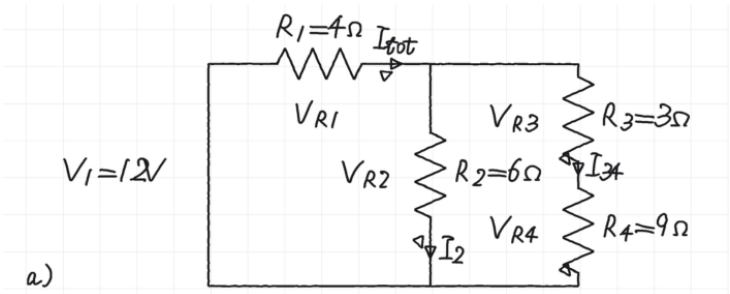
- Resolution checks do not prove exact topology. Finite domains and sampling can miss small features.
- Results depend on the domain, sampling resolution and filtration scale.

Scrawl AI

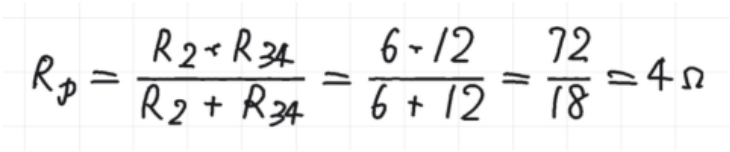
Sole creator, developer and operator | Launched 2026 | Approximately 300 paying customers

Scrawl extracts your handwriting from photos or PDFs of your work, solves uploaded problems, and returns worked solutions in your own handwriting. I built the extraction and rendering engines, web/iOS apps and paid service.

CIRCUIT OUTPUT / DETAIL



HANDWRITTEN EQUATION / DETAIL



Scrawl demo output: circuit and handwritten equation details; not customer data.

Handwriting extraction

- Detected text lines with PaddleOCR, split glyphs with OpenCV and classified crops with Gemini.
- Saved the samples in editable, per-user handwriting libraries.

Handwriting renderer

- Used LaTeX/PyMuPDF layout to position custom glyphs in text and equations.
- Modeled correlated variation in slant, spacing and baselines, with word-aware glyph selection.

Math and diagram layout

Reconstructed stretched brackets and fraction bars; redrew supported vector paths while keeping diagram labels aligned. Added missing-glyph fallbacks and quality warnings.

Product ownership

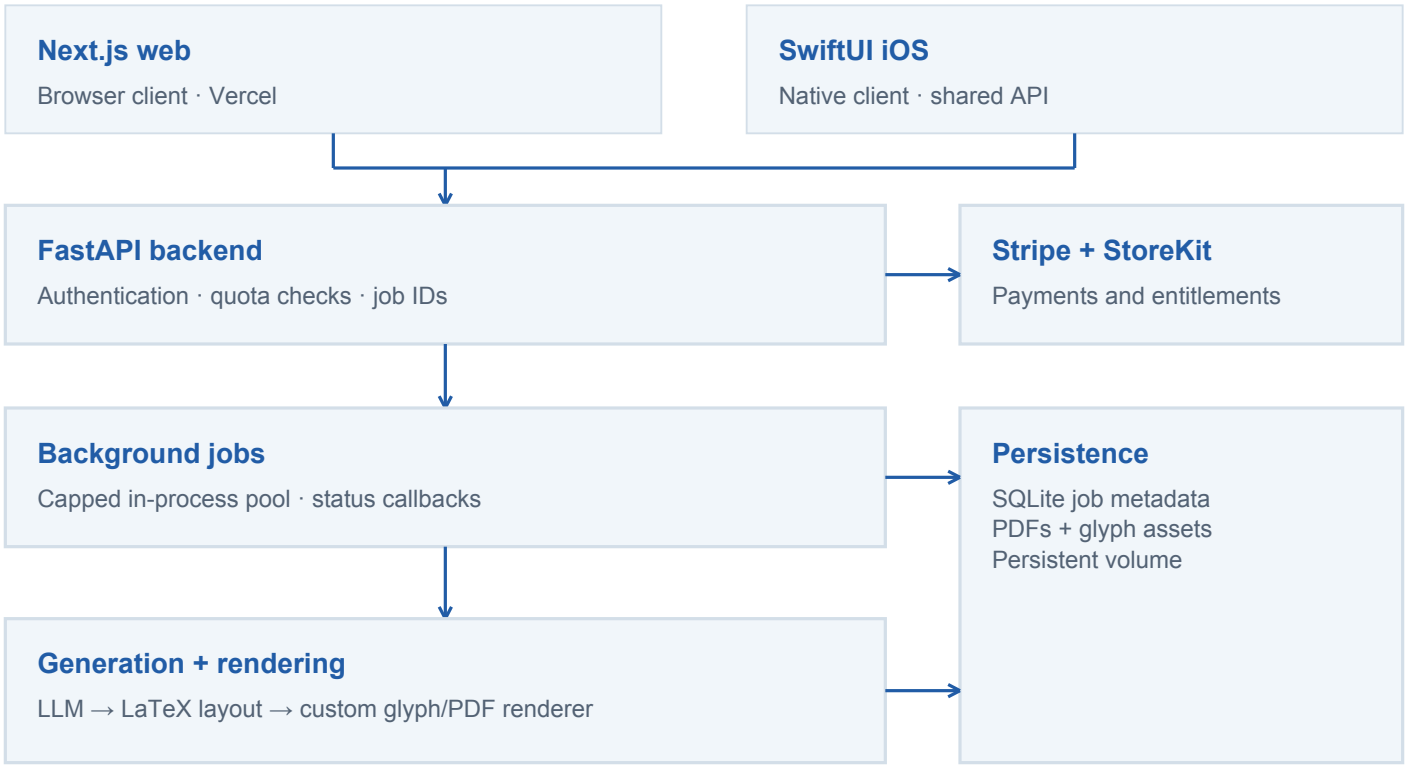
Built and operated the service, set up hosting and ran advertising, from launch through ongoing product support.

scrawlai.app

Scrawl AI architecture

Application, generation interface and product discovery

Built Next.js and SwiftUI clients on a shared FastAPI backend, with asynchronous generation, saved handwriting libraries and subscriptions.



Clients receive a job ID, poll status and download PDFs. Jobs run in a capped in-process pool; interrupted jobs require recovery after a backend restart.

Generation interface

- Showed generation stages in iOS, with completion notifications and PDF previews.
- Added background glyph uploads and saved drawing drafts to reduce interruptions.

Search and AI discovery (GEO)

- Added crawlable subject/sample pages, JSON-LD, sitemaps and RSS.
- Published llms.txt references and explicit AI-crawler rules.

Subscription state

Integrated Stripe/StoreKit entitlements with atomic credit reservations and idempotent purchase grants to prevent duplicate billing-state updates.

Costream

Member of Technical Staff | Dec 2024–Present | Hybrid
Formerly CandyTrail | Y Combinator Summer 2025

One of Costream's first five engineers. Worked on Python backend development, audio AI research data and customer deployments.

My work

- Contributed to Python backend development and optimized pipelines that processed 50,000+ customer interactions.
- Managed technical onboarding for three enterprise pilots, resolved deployment issues and incorporated customer feedback.
- Acquired research data for audio AI development and negotiated with external vendors.

Results

50,000+ customer interactions processed by the pipelines.

3 enterprise pilots onboarded.

Current project details are confidential.

Company context: Y Combinator Summer 2025.

Costream on Y Combinator

gzhang45@jh.edu